

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems

COURSE CODE: CSA0501

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COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Technical Presentation (Low)

Assessment Tool Weightage: 35%

Dr

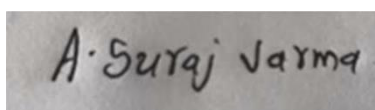
QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.	BL2 – Understand	5
2	Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.	BL2 – Understand	5
3	Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.	BL2 – Understand	5
4	Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.	BL1 – Remember	5
5	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand	5
6	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand	5
7	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand	5

8	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember	5
9	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand	5
10	Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.	BL2 – Understand	5

Code of Conduct:

I A. Suraj varma (192525339) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach

Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

INTRODUCTION TO DBMS

- **What is DBMS?**
- A **Database Management System (DBMS)** is software used to store, organize, retrieve, and manage data efficiently.
- It acts as an interface between users and databases.
- DBMS helps in storing large amounts of data securely.
- It provides easy access to information whenever required.
- It ensures data accuracy, consistency, and security.
- DBMS supports multiple users accessing the same database at the same time.

COMPARISON – FILE SYSTEM VS DBMS

Click to add text

File System

Data stored in separate files
High data redundancy
Less secure
Difficult backup
No relationships between files
Slow data searching
Limited multi-user support

DBMS

Data stored in a database
Low data redundancy
Highly secure
Easy backup & recovery
Supports relationships between tables
Fast querying and retrieval
Supports multiple users

ANSI/SPARC ARCHITECTURE

ANSI/SPARC Architecture

ANSI/SPARC Architecture is a **three-level database architecture** used in DBMS. It separates the database into **External Level, Conceptual Level, and Internal Level**.

External Level: Provides different views of the database for different users.

Conceptual Level: Describes the complete logical structure of the database. It provides **data independence**, improves security, and supports multiple users.

Practical Example

Bank Management System

Customer → Views account details.

Bank Employee → Accesses customer records.

DBA → Manages database storage and security.

DATA MODELS

- **Data Models in DBMS**

- A **Data Model** is a collection of concepts used to describe the structure of a database.
- It defines how **data is stored, organized, and related** within the database.
- Data models help in designing databases efficiently and maintaining data consistency.
- **Hierarchical Model:** Organizes data in a tree structure with parent-child relationships.
- **Network Model:** Allows a record to have multiple parent and child records using a graph structure.
- **Relational Model:** Stores data in tables (rows and columns) and is the most widely used model.
- **Entity-Relationship (ER) Model:** Represents data using entities, attributes, and relationships during database design.

ER DIAGRAM (ONLINE LIBRARYMANAGEMENTSYSTEM)

- **Entity-Relationship (ER) Diagram – Online Library**
- An **ER Diagram** is used to represent the **entities, attributes, and relationships** in a database.
- It helps in designing the database before implementation.
- The main entities in an **Online Library** are **Member, Book, Librarian, and Issue**.
- A **Member** can borrow one or more **Books**.
- A **Librarian** manages books and issues them to members.
- The **Issue** entity stores issue date, return date, and fine details.
- ER diagrams improve database organization and reduce data redundancy.

DATA INDEPENDENCE

- **Data Independence**
- **Data Independence** is the ability to modify the database structure **without affecting application programs**.
- It allows changes at one level of the database without changing the other levels.
- It makes the database more flexible and easier to maintain.
- It reduces the cost and effort required for database modifications.

Data Independence is one of the main advantages of the **ANSI/SPARC three-level architecture**.

Types of Data Independence

- **1. Physical Data Independence:** Changes in the **physical storage** (such as indexing or file organization) do not affect the logical structure or application programs.
- **2. Logical Data Independence:** Changes in the **logical structure** (such as adding new attributes or tables) do not affect the user views or application programs.

ENHANCED ENTITY-RELATIONSHIP (EER) MODEL

The **Enhanced Entity-Relationship (EER) Model** is an advanced version of the ER Model.

It provides additional features to represent complex database structures.

EER supports **Specialization**, where one entity is divided into sub-entities based on characteristics.

It supports **Generalization**, where multiple similar entities are combined into one generalized entity.

It also supports **Inheritance**, allowing sub-entities to inherit attributes from the parent entity.

EER helps in designing large and complex database systems more efficiently.

It improves database organization, flexibility, and accuracy.

SOFTWARE COMPONENTS OF DBMS

Software Components of DBMS

A **Database Management System (DBMS)** consists of several software components that work together to manage data efficiently.

Query Processor: Interprets and executes SQL queries submitted by users.

Storage Manager: Manages data storage, retrieval, indexing, and file organization.

Transaction Manager: Ensures that all database transactions are completed correctly while maintaining consistency.

Recovery Manager: Restores the database after system failures using backup and recovery techniques.

Authorization & Integrity Manager: Controls user access permissions and maintains data security and integrity.

These components improve the **performance, security, reliability, and efficiency** of the database system.

STRONG ENTITY VS WEAK ENTITY

- **Strong Entity and Weak Entity**
- An **Entity** is a real-world object that stores information in a database.
- A **Strong Entity** has its own **Primary Key** and can exist independently.
- A **Weak Entity** does **not** have a Primary Key and depends on a Strong Entity for its existence.
- A Weak Entity is identified using a **Partial Key** along with the Primary Key of the Strong Entity.
- **Strong Entity** is represented by a **single rectangle** in an ER diagram.
- **Weak Entity** is represented by a **double rectangle**, and the identifying relationship is shown using a **double diamond**.
- Strong and Weak Entities help in designing accurate and efficient database systems.

ENHANCED ENTITY-RELATIONSHIP (EER) DIAGRAM FOR RETAIL CHAIN MANAGEMENT

- **EER Diagram – Retail Chain Management**
- An **Enhanced Entity-Relationship (EER) Diagram** represents entities, relationships, specialization, and inheritance in a database.
- In a **Retail Chain Management System**, the main entities are **Store, Employee, Customer, Product, Order, and Supplier**.
- A **Customer** places one or more **Orders**.
- Each **Order** contains one or more **Products**.
- A **Supplier** supplies products to different **Stores**.
- **Employee** can be specialized into **Manager** and **Sales Executive** using the **ISA (is-a)** relationship.
- The EER model improves database organization, reduces redundancy, and supports complex business operations.